Electrically written quasi-phase matching of intersubband second-harmonic generation: bias-reversed asymmetric and mirror-symmetric quantum wells

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Abstract

Resonant intersubband nonlinearities in quantum wells exceed those of bulk crystals by orders of magnitude, but in waveguides they are limited by phase mismatch, resonant absorption and saturation. Quasi-phase matching (QPM) written by the bias of segmented electrodes was proposed in the patent literature, but it has not been established whether such a scheme can compete with growth-defined orientation reversal or with Stark on/off QPM. We compare two mechanisms for bias-defined QPM. In an ordinary asymmetric double quantum well (ADQW), bipolar bias Stark-shifts the 1–2 transition to either side of the pump, which reverses the phase of the resonant $\chi^{(2)}$ (151–189° between segments) while detuning both segments, so they saturate less. In a mirror-symmetric coupled triple well, the field localizes the ground state in one outer well, so the structure behaves as an ADQW whose orientation follows the field; $\chi^{(2)}$ is exactly odd and the linear response exactly even in the field, so alternating polarity writes an exact sign grating. For $In_{0.53}Ga_{0.47}As/In_{0.52}Al_{0.48}As$ structures pumped at $10\,\mu$m, self-consistent Schrödinger–Poisson calculations with depolarization give the triple well (58/14/34/14/58 Å) $|\chi^{(2)}| = 61$ nm/V at 55 kV/cm, 55% of an optimized ADQW, robust to model, material and linewidth variations; symmetric single, double and stepped wells reach 34–40% only at 140 kV/cm, the highest field at which our single-period model is valid. In an InP waveguide, segmented electrodes at 30 V retain 92% of the ideal sign-reversal coefficient, but no single factor describes the electrodes: above 1 W the fringing fields change the efficiencies by -44% to $+38\%$ relative to abrupt fields, depending on scheme and power. Coupled-wave simulations with LO-phonon-resolved multilevel saturation and pump depletion propagate every electrically written pattern through the field its electrodes produce, with a lateral breakdown limit between neighbouring electrodes and every field restricted to where a single-period model is valid. They give the following. The ADQW under bipolar bias ($\pm$55–90 kV/cm) is the most efficient scheme above 1 W, by 1.4–2.4 times over the bias-optimized triple well. The triple well in turn gives 1.9–2.4 times the conversion of Stark on/off QPM at 1.26 W and 1.2–1.8 times at 10 W, and 1.9–9.7 times that of an ADQW with growth-defined orientation reversal at 1.26–20 W. At 10 mW all electrical schemes are within a factor of 1.3 of each other. Absolute efficiencies (0.11–0.75% at 10 W for the triple well at its design field) depend on the relaxation pathways. Bias-defined phase reversal near resonance therefore does not need mirror symmetry, and the bipolar ADQW is the choice for efficiency; the triple well offers an exact sign grating with identical segments, no absorption or index grating at the QPM period and a bias-independent period. We give a growth- and fabrication-ready triple-well design, its operating constraints, and the measurements that would test it.

1 Introduction

Intersubband (ISB) transitions in doped semiconductor quantum wells (QWs) can be engineered to place two or three transitions in resonance with a pump and its harmonic, giving second-order susceptibilities $\chi^{(2)}$ of order 10^4–10^6 pm/V in the mid-infrared [1–6]. These nonlinearities underpin difference-frequency terahertz sources integrated into mid-infrared quantum cascade lasers (QCLs) [7, 8] and nonlinear polaritonic metasurfaces [9–13]. In waveguides, however, the resonant enhancement comes with strong absorption at both frequencies, saturation at MW/cm^2 intensities [14], and a coherence length $L_{\mathrm{coh}} = \pi/\Delta k$ of only tens of micrometres, so the useful interaction length is short and the conversion efficiency is low [5, 6, 15].

Several routes to phase matching have been explored in QW structures. True phase matching has been obtained by adding QWs whose resonant dispersion compensates the material and waveguide dispersion [5, 16], and Meyer *et al.* proposed imposing it electrically through the voltage dependence of that resonant index [17]; Vurgaftman *et al.* showed that a uniform field across an InGaAs/InAlAs ADQW waveguide moves the resonant contribution to the index mismatch enough to switch a 50-μm, 14% converter off at -25 kV/cm, or to keep at least 1% conversion from 9 to 20 μm pump wavelength; there the field is uniform and acts on the coherence length, not on the sign of $\chi^{(2)}$, and they proposed separating the active and phase-matching regions [18, 19]. Quasi-phase matching (QPM) has been implemented by growing asymmetric coupled wells with periodically reversed orientation [20, 21], by spatially selective intermixing that switches $\chi^{(2)}$ off in alternate domains [22, 23], and by periodically Stark-detuning ISB resonances along a QCL ridge [24]. Magnitude modulation is weaker than sign reversal: an ideal on/off pattern has a first Fourier coefficient of $1/\pi$ against $2/\pi$ for sign reversal, a factor of four in conversion efficiency. Stark detuning also changes the linear absorption and refractive index from segment to segment. Near resonance, however, a Stark shift that moves a transition across the pump also reverses the phase of $\chi^{(2)}$. Meyer *et al.* placed the 1–2 transition of an ADQW at the pump so that reversing the bias polarity changes the resonant index strongly, and used this for phase matching with a uniform field [17]; we use the same crossing to reverse the phase of $\chi^{(2)}$ from segment to segment, and show in Secs. 4.6 and 4.7 that this route, with bipolar bias on an ordinary ADQW, is the most efficient electrical pattern we have found. Bias control of this detuning phase has been demonstrated in intersubband polaritonic metasurfaces [25]; here we apply it along a waveguide to quasi-phase-match. Orientation reversal by growth fixes the period at epitaxy and requires regrowth or patterned growth. Outside QWs, field-induced $\chi^{(2)} = 3\chi^{(3)}E$ is sign-reversed with alternating junctions in silicon [26] and programmed optically in silicon nitride [27], but the induced nonlinearity is small (0.5–41 pm/V) and needs tens to a thousand volts.

That a perpendicular field induces a $\chi^{(2)}$ in a symmetric QW whose sign follows the field was predicted [28] and observed [1]. A 1989 patent proposed spatially periodic biasing of such wells in a waveguide for QPM [29], and a 1994 patent described $+\chi^{(2)}$ and $-\chi^{(2)}$ zones written by the direction of the field in a symmetric (stepped) well [30]. We are not aware of a quantitative design of this scheme, a comparison with established QPM methods and with other bias patterns, or a demonstration. The obstacle is magnitude: in a symmetric single or stepped well, $\chi^{(2)}$ grows only gradually with field and remains well below that of an ADQW. Here we ask whether QPM written by the bias of segmented electrodes can compete with growth-defined orientation reversal and with Stark on/off QPM, and we examine two routes: bipolar bias on an ordinary ADQW, which reverses the phase of the resonant $\chi^{(2)}$ by Stark detuning, and a mirror-symmetric coupled triple well, which removes the magnitude obstacle of symmetric wells and writes an exact sign grating. Our contributions are: (i) a comparison of the electrical schemes with growth-defined orientation

reversal, Stark on/off QPM and phase matching in a common waveguide, electrode and saturated coupled-wave model with LO-phonon-resolved relaxation, including fully optimized two-field bias patterns, which shows that bias-defined QPM is competitive above about 1 W and that the bipolar ADQW is the most efficient electrical scheme; (ii) the first route, bipolar bias on an ADQW, with the phase reversal by detuning quantified at the optimum fields and its costs (unequal segments, an absorption and index grating at the QPM period) identified; (iii) the second route, a mirror-symmetric triple well that acts as an *electrically reversible* ADQW, reaching about half of an optimized ADQW's $\chi^{(2)}$ at 55 kV/cm, whereas symmetric single, double and stepped wells reach 34–40% only at 140 kV/cm, the highest field at which our model is valid, together with an exact symmetry argument showing that bias reversal writes a pure sign grating, with the residual linear modulation quantified; (iv) a self-consistent analysis, including the electrons' own space charge and depolarization, with an explicit error budget; and (v) a design, its operating constraints and a test plan specific enough to be grown and measured.

2 Concept

Let Π be the mirror operation $z \to -z$ about the centre of a structure whose layer sequence is symmetric. The Hamiltonian with a uniform field F along z satisfies $\Pi H(F)\Pi^{-1} = H(-F)$. Its eigenstates at $-F$ are therefore the mirror images of those at $+F$, with identical energies and with dipole matrix elements $z_{ij}(-F) = -z_{ij}(F)$ (up to the arbitrary signs of the states, which cancel in closed loops). Every term of the second-order susceptibility contains a closed loop of three dipoles, $z_{lm}z_{mn}z_{nl}$, so

$$\chi^{(2)}(-F) = -\chi^{(2)}(F), \qquad \chi^{(2)}(0) = 0, \tag{1}$$

while every linear quantity, which contains $|z_{lm}|^2$, is even in F. A periodic electrode pattern of alternating polarity therefore produces a pure sign grating of $\chi^{(2)}$: oppositely biased segments have identical linear absorption and refractive index (the gaps between electrodes are treated in Sec. 4.4). The same argument holds when the electrons' own Hartree potential and depolarization are included, provided the doping is placed mirror-symmetrically.

The open question is magnitude. A symmetric single well acquires $\chi^{(2)}$ only linearly in F at small fields [1]. A coupled triple well (Fig. 1a) behaves differently. Its two outer wells form a near-degenerate pair (splitting 11 meV at zero field), so a modest field localizes the ground state in one outer well, and the structure then resembles an ADQW of outer well, barrier and centre well. The polarity of F selects which outer well, and hence the orientation of the effective ADQW. Fig. 1b shows the device: a waveguide containing a stack of such triple wells, with segmented top electrodes of alternating polarity over a common ground contact.

3 Model

3.1 Electronic structure and nonlinear susceptibility

We use a single conduction band with the BenDaniel–Duke kinetic operator [31] and Kane-type nonparabolicity, $m^*(z,E) = m_0^*(z)\,[1 + (E - V(z))/E_g(z)]$ [32], for $In_{0.53}Ga_{0.47}As$ wells ($m_0^* = 0.043$, $E_g = 0.75$ eV) and $In_{0.52}Al_{0.48}As$ barriers ($m_0^* = 0.075$, $E_g = 1.45$ eV) with a 0.52 eV conduction-band offset [33]. The Schrödinger equation is discretized on a cell-centred 0.5 Å grid, which makes mirror-symmetric stacks

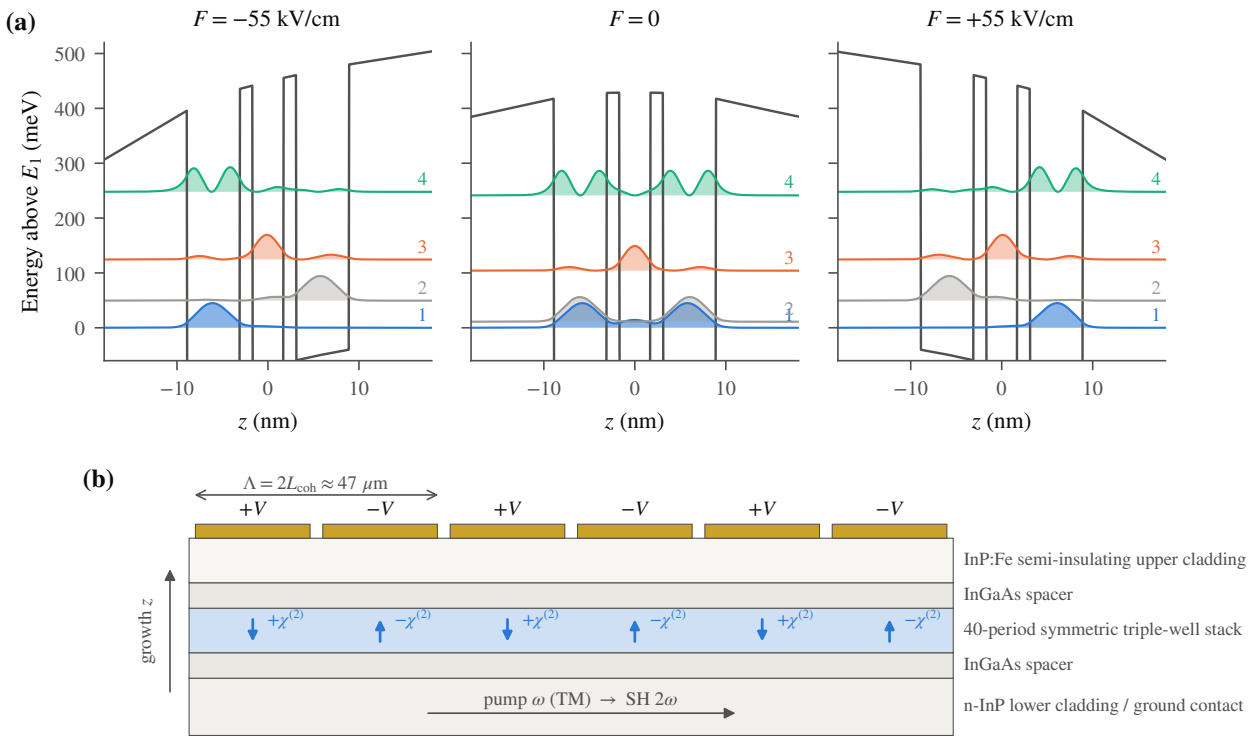


Figure 1: (a) Self-consistent conduction-band profile and probability densities of the lowest four subbands of the 58/14/34/14/58 Å InGaAs/InAlAs triple well at −55, 0 and +55 kV/cm (300 K, 5×10^{11} cm^{-2} per period). At zero field the ground state is shared by the two outer wells and $\chi^{(2)} = 0$; a field localizes it in one outer well, forming an effective ADQW whose orientation follows the field. (b) Device concept: segmented electrodes of alternating polarity over a grounded n-InP contact reverse the sign of $\chi^{(2)}$ every coherence length.

exactly symmetric on the grid. Each state is found self-consistently with its own energy-dependent mass by requiring that it remain the k-th eigenvalue of $H(E)$; this preserves node counting and avoids state swapping. The resulting states are not exactly orthogonal and are Löwdin-orthonormalized [34] together with the states of the computational box above them, which makes the dipoles independent of the box. Using instead the two-band Kane states, which are orthogonal without the Löwdin step (overlaps between localized states below 10^{-4}), changes $|\chi^{(2)}|$ at the pump by -0.9% for the ADQW used below (64/10/29 Å) and by -0.7% for the triple well at 55 kV/cm. Only states with at least 80% of their probability in the active region are kept.

The resonant SHG susceptibility for arbitrary subband populations ρ_l is the density-matrix result [2, 35]

$$\chi^{(2)}(2\omega;\omega,\omega) = \frac{Ne^3}{\varepsilon_0} \sum_{l,m,n} \frac{z_{lm} z_{mn} z_{nl}}{\tilde{E}_{nl} - 2\hbar\omega - i\Gamma} \left[\frac{\rho_l - \rho_m}{\tilde{E}_{ml} - \hbar\omega - i\Gamma} - \frac{\rho_m - \rho_n}{\tilde{E}_{nm} - \hbar\omega - i\Gamma} \right], \tag{2}$$

where N is the electron density averaged over a period, $\Gamma = 10$ meV is the half-width at half-maximum of each transition, and $\tilde{E}_{ml}$ are depolarization-shifted transition energies (Sec. 3.2). Eq. (2) is invariant under a shift of the coordinate origin, contains the permanent-dipole term $z_{12}^2(z_{22} - z_{11})$ automatically, and reduces to the three-level expression of Ref. [6] when only the ground state is occupied. Populations follow Fermi–Dirac statistics at 300 K for a sheet density n_s per period. We compare designs at fixed $n_s = 5 \times 10^{11}$ cm^{-2} per period with a 200 Å $In_{0.52}Al_{0.48}As$ spacer, so that both structure types hold the same number of electrons per period.

3.2 Self-consistent electrostatics and depolarization

Donors are placed as sheets at the centres of the spacers (mirror-symmetric). The stack is voltage-biased, so the period-averaged field equals the applied F and the Hartree field has zero mean over a period. We solve Poisson's equation periodically with position-dependent permittivity (13.9 for InGaAs, 12.4 for InAlAs) and iterate to self-consistency, to 10^{-5} eV in the potential. Depolarization shifts follow Ando *et al.* [36], applied transition by transition with an average permittivity $\varepsilon = 13.5$:

$$\tilde{E}_{ml}^2 = E_{ml}^2 \left[1 + \frac{2e^2 n_s (\rho_l - \rho_m) S_{lm}}{\varepsilon \varepsilon_0 E_{ml}} \right], \quad S_{lm} = \int dz \left[\int_{-\infty}^{z} \psi_l \psi_m \, dz' \right]^2 . \tag{3}$$

Exciton-like corrections, coupling between transitions, and the local-field correction of the driving field [37, 38] are not included.

3.3 Waveguide, electrodes and propagation

TM modes of the planar waveguide are computed by finite differences for H_y with complex permittivities. InP follows the Pettit–Turner dispersion [39]; InGaAs and InAlAs are taken as $n = 3.43$ and 3.20 at $10\,\mu$m with the same dispersion as InP; n-InP is described by a Drude term; and gold electrodes use Ordal's Drude parameters [40]. The stack's z-polarized response is entirely its ISB response and is treated in the propagation model rather than as free-carrier absorption.

The field under the electrodes is the Fourier-series solution of Laplace's equation between a segmented top plane (segments $\pm V$ separated by gaps with linearly varying potential) and a grounded bottom contact, for a uniform permittivity (the layers differ by less than 12%). Local $\chi^{(2)}(F(x))$ is taken from the self-consistent $\chi^{(2)}(F)$ curve, and the QPM coefficient is the first Fourier component $c_1 = |\Lambda^{-1} \int_0^\Lambda \chi^{(2)}(x) e^{-2\pi i x/\Lambda} dx|$, equal to $(2/\pi)|\chi^{(2)}|$ for an ideal sign-reversed square wave. This coefficient characterizes the electrodes, but we do not use it as a factor: every electrically written pattern is propagated through the field that its electrodes produce. For an arbitrary sequence of electrode voltages, Laplace's equation in the slab is solved exactly by Fourier transform, and the field is evaluated at the bottom, centre and top of the QW stack; each height takes the self-consistent medium (with its own $\chi^{(2)}$, absorption, ISB index and relaxation, below) of the nearest valid bias on a 5 kV/cm grid (fields where the single-period model is not valid, Sec. 4.6, take the nearest valid one), each height keeps its own populations, and $\chi^{(2)}$, the absorption and Δk are averaged over the three with weights $1/4$, $1/2$, $1/4$ [41]. The field between electrodes at different voltages therefore passes through the intermediate biases, with their absorption and phase. As elsewhere in this work, every segment is one coherence length of its own medium and Δk is that of the equilibrium populations (the cited field model can also follow the saturation of the ISB index, which we do not include here). Neighbouring electrodes also put a lateral field across the gap; we keep it below 0.5 MV/cm, about the breakdown field of InP, so that with $2\,\mu$m gaps neighbouring segments may differ by at most 188 kV/cm (100 V).

Propagation uses mode amplitudes $a_{1,2}$ ($|a|^2$ is power) generalizing Eq. (15.29) of Ref. [6]:

$$\frac{da_1}{dx} = -\frac{\alpha_1}{2} a_1 - iK\, s(x)\, \chi^{(2)*} a_2 a_1^* e^{-i\Delta k x}, \tag{4}$$

$$\frac{da_2}{dx} = -\frac{\alpha_2}{2} a_2 - iK\, s(x)\, \chi^{(2)} a_1^2 e^{+i\Delta k x}, \qquad K = \frac{\pi}{\lambda_1} \sqrt{\frac{2\mathscr{O}}{n_1^2 n_2 c \varepsilon_0}}, \tag{5}$$

where $\mathcal{O} = |\int_{\mathrm{QW}} f_1^2 f_2^* \, dz|^2/w$ is the nonlinear overlap of the normalized TM fields $f_{1,2}$ for a ridge of width $w = 15\,\mu$m, and $s(x)$ is the QPM pattern. The modal losses are $\alpha_j = \Gamma_j \alpha_{\mathrm{ISB},j}(\rho) + \alpha_{\mathrm{bg},j}$. Saturation enters through steady-state rate equations for all subband populations,

$$0 = \sum_l W_{lm}(\rho_l - \rho_m) + \sum_l \left[R_{lm}(\rho_l - \rho_l^{(0)}) - R_{ml}(\rho_m - \rho_m^{(0)}) \right], \tag{6}$$
$$W_{lm} = \frac{\sigma_{lm}(\omega) I_1}{\hbar\omega} + \frac{\sigma_{lm}(2\omega) I_2}{2\hbar\omega},$$

with per-electron cross-sections σ_{lm} from the depolarization-shifted Lorentzian transitions, local intensities $I_j = P_j \Gamma_j/(t_{\mathrm{stack}} w)$, and equilibrium (Fermi–Dirac) populations $\rho^{(0)}$. The relaxation rates R_{lm} from subband l to m are computed for each structure from the Fröhlich interaction with bulk LO phonons [42],

$$R_{lm} = \frac{e^2 m^* \omega_{\mathrm{LO}}}{4\varepsilon_0 \hbar^2} \left(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_s} \right) \left[(n_{\mathrm{LO}} + 1) \frac{I(Q_e)}{Q_e} + n_{\mathrm{LO}} \frac{I(Q_a)}{Q_a} \right], \tag{7}$$
$$I(Q) = \iint \psi_l(z) \psi_m(z) \, \psi_l(z') \psi_m(z') \, e^{-Q|z-z'|} \, dz \, dz',$$

for electrons at the bottom of subband l with $E_l > E_m$, where $\hbar^2 Q_{e,a}^2/2m^* = E_l - E_m \mp \hbar\omega_{\mathrm{LO}}$ (emission only if positive) and n_{LO} is the 300 K phonon occupation; upward rates follow from detailed balance. We use $\hbar\omega_{\mathrm{LO}} = 33.7$ meV, $\varepsilon_\infty = 11.6$, $\varepsilon_s = 13.9$ and $m^* = 0.043$ for $In_{0.53}Ga_{0.47}As$. Interface-roughness, alloy and electron–electron scattering are neglected; we test their effect by adding a 1 ps channel to the ground state, and we repeat the comparison with a single relaxation time $\tau = 1$ ps for all transitions [6]. Because $\chi^{(2)}$ and α_{ISB} in Eq. (2) are linear in ρ, both are recomputed exactly at every step. The phase mismatch includes the modal dispersion and each medium's own ISB contribution to the refractive index. For segmented schemes in which the medium changes from segment to segment (Stark on/off QPM), each segment has its own $\chi^{(2)}$, absorption, relaxation and phase mismatch, and its length equals its own coherence length.

3.4 Validation

All numbers come from the open-source package `qwkit` (v0.7.0), archived with the scripts and data for every figure and table (see Data and code availability). Its 69 automated tests include those summarized in Table 1. The LO-phonon benchmark agrees to 13%, within the uncertainty from the unstated ε_∞ of the reference calculation. The symmetry tests are the most demanding: earlier versions of the solver produced a small spurious $\chi^{(2)}$ at zero field, from grid asymmetry, state swapping and non-orthogonality, and the tests were written to catch exactly these errors.

4 Results

4.1 Symmetric designs against an optimized ADQW

We optimized an ADQW (wide well, barrier, narrow well) and four symmetric families (single, double, triple and stepped wells) for a $10\,\mu$m pump at fixed n_s, first with a fixed potential and then self-consistently. The best ADQW is 64/10/29 Å, with $|\chi^{(2)}| = 110.9$ nm/V in the full model. Fig. 2a shows the best symmetric designs as a function of field. The triple well rises steeply, reaching 55% of the ADQW at 55 kV/cm; the best triple well found in the self-consistent search (59/12/36 Å) reaches 57% at 65 kV/cm, and we retain

Table 1: Validation tests in `qwkit` v0.7.0 (all pass).

Test	Criterion
Infinite-well limit	energies within 1%, z_{12} within 2% of analytic
Book Eq. (15.28) three-level limit	Eq. (2) within 5%
Symmetric wells (fixed and self-consistent potential)	$\chi^{(2)}(0) = 0$ and $\chi^{(2)}(-F) = -\chi^{(2)}(F)$ to 10^{-6} relative
Translation invariance	$\chi^{(2)}$ unchanged by origin shift to 10^{-9}
State parity with nonparabolicity	exact parity, strictly ordered energies
Charged-sheet electrostatics	potential within 1% of analytic
Self-consistency	charge neutrality $< 10^{-6}$; convergence $< 10^{-5}$ eV
Depolarization	blue shift for populated ground state; vanishes as $n_s \to 0$
Symmetric dielectric slab, TM_0	n_{eff} within 2×10^{-4} of analytic
Parallel-plate electrodes	field within 1% of V/d
QPM coefficients	$2/\pi$ (sign reversal) and $1/\pi$ (on/off) within 10^{-3}
Undepleted, lossless, phase-matched SHG	$P_2 = \eta_n P_1^2 L^2$ within 10^{-3}
Lossless depletion	$P_1 + P_2$ conserved to 10^{-6}
Two-level saturation	population difference and absorption halve at I_{sat}
LO-phonon form factor $I(Q)$	direct double sum to 10^{-9}
Form factor, short-range limit	$I(Q) \to (2/Q) \int \psi_l^2 \psi_m^2 dz$ within 1%
LO-phonon benchmark (10 nm GaAs well [43])	E_{21} within 0.3 meV; $\tau_{21} = 0.73$ vs 0.83 ps
Detailed balance	Boltzmann populations stationary to 10^{-6}
Level-resolved rate equations	equilibrium stationary; populations conserved
Segmented propagation	reproduces sign-reversal QPM to 10^{-6}

58/14/34 Å for its lower operating field. Up to 140 kV/cm, the highest field at which we consider the single-period model valid (Sec. 4.6), the symmetric double and single wells reach only 39% and 40%, both at that limit. Their $|\chi^{(2)}|$ keeps rising at higher fields (to 90% for the double well at 200 kV/cm), but there their third subband lies above the barrier band edge at the period boundary, where a single-period model no longer describes it. We also optimized the symmetric stepped well of Ref. [30], an InGaAs core between $In_{0.53}(Ga_{1-x}Al_x)_{0.47}As$ steps inside InAlAs barriers, over core width (20–80 Å), step width (10–60 Å), step composition ($x = 0.2$–0.5, parameters interpolated linearly between the ternaries) and field. The design shown in Fig. 2a, 50/40/50 Å with $x = 0.35$, reaches 25% at 140 kV/cm, and no refined candidate exceeds 34% up to that field; higher values (up to 48% at 180 kV/cm) again occur only beyond it.

Fig. 2b shows the mechanism. At zero field the outer-well doublet is split by 11 meV and $z_{13} = 0$ by parity. With field, the doublet splits and the ground-state centroid moves to $|\langle z \rangle_1| = 5.7$ nm at 55 kV/cm, close to the outer-well centre at 6.0 nm. At 55 kV/cm the ladder $1 \to 3 \to 4$ is doubly resonant ($E_{31} = 124.4$ meV, $E_{41} = 247.8$ meV against $\hbar\omega = 124.0$ meV), with $z_{13} = 1.24$, $z_{34} = 2.40$ and $z_{14} = 1.12$ nm.

4.2 What the self-consistent physics changes

Table 2 shows the triple-well-to-ADQW ratio at increasing levels of model and for different donor placements. The electrons in the triple well move by ∼5.7 nm under bias, but because each period is voltage-clamped the space charge only redistributes the field inside the period: the optimum field shifts from 50 to 55 kV/cm and the ratio stays between 55% and 63%. Over $n_s = 10^{10}$–5×10^{11} cm^{-2} the ratio is 55–61%.

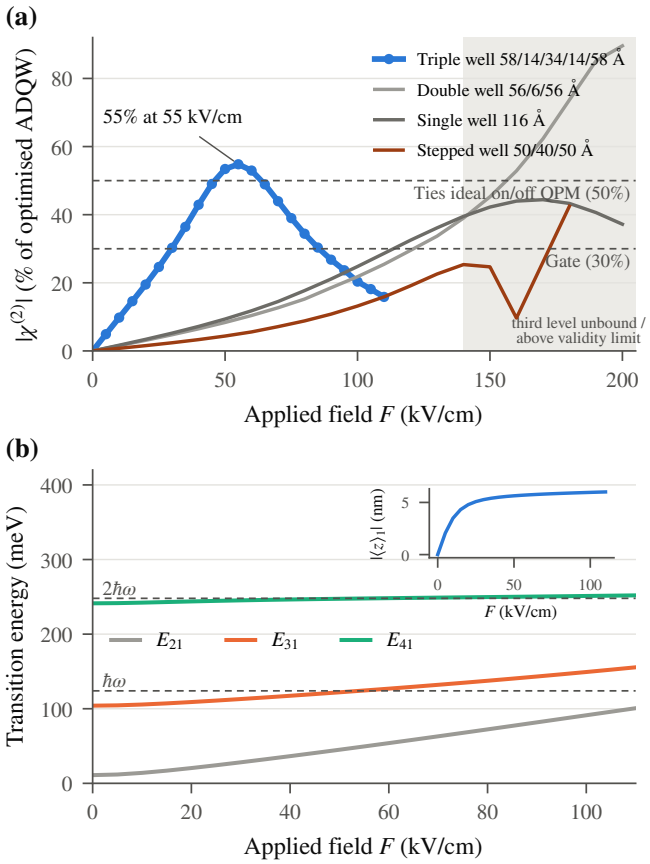


Figure 2: (a) $|\chi^{(2)}|$ of the best symmetric designs as a percentage of the optimized ADQW, versus applied field, full self-consistent model (300 K, $n_s = 5 \times 10^{11}$ cm^{-2}, $\Gamma = 10$ meV, 10 μm pump). The shaded region above 140 kV/cm, where the third level lies above the barrier edge at the period boundary, is beyond the range in which the single-period model is valid (Sec. 4.6); the stepped well ($In_{0.53}(Ga_{0.65}Al_{0.35})_{0.47}As$ steps) has fewer than three localized levels at 170 kV/cm, where its point is omitted, and is shown up to 180 kV/cm. (b) Transition energies of the triple well versus field, with the pump and SH photon energies; inset, ground-state centroid relative to the structure centre.

4.3 Robustness

Fig. 3 re-evaluates the ratio with the full model under parameter and growth errors, re-tuning the triple well's field in each case (the field is a free parameter of the device) and keeping the ADQW at zero bias. Band-offset ($\pm$0.02 eV), effective-mass ($\pm$5%) and linewidth (6–14 meV) variations keep the ratio at 53–62%; the linewidth changes both structures almost equally. A one-monolayer ($\pm$3 Å) error in the outer wells is the most damaging for the triple well (36–38%), because those wells set the doublet on which the mechanism relies. The same errors in the ADQW at fixed bias move the ratio to 74–112%. A biased ADQW could partly re-tune, so the fair conclusion is that both designs require monolayer control, with the triple well's bias acting as a built-in trim.

4.4 Waveguide and electrodes

The waveguide (Fig. 4a) has, from the top, gold electrodes on 2.0 μm of semi-insulating InP:Fe, a 0.9 μm InGaAs spacer, a 40-period stack (1.51 μm), a second 0.9 μm InGaAs spacer and an n-InP (10^{17} cm^{-3}) lower

Table 2: Triple well (58/14/34 Å) against the ADQW (64/10/29 Å) at different model levels; each design at its optimum field.

Model	Triple well (nm/V)	Field (kV/cm)	ADQW (nm/V)	Ratio
Fixed potential, no depolarization	61.2	50	105.7	58%
Fixed potential, depolarization	59.3	45	93.8	63%
Hartree, no depolarization	60.7	55	103.2	59%
Hartree + depolarization (full)	**60.8**	**55**	**110.9**	**55%**
Full, donors in all wells	60.1	60	105.0	57%
Full, donors in the centre well	58.7	70	105.0	56%

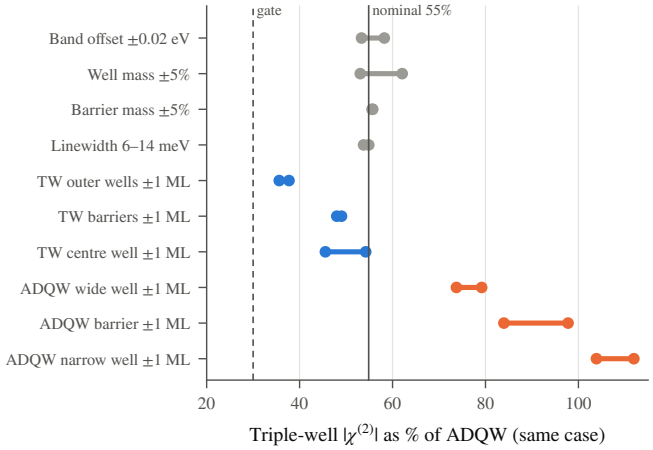


Figure 3: Error budget: triple-well $|\chi^{(2)}|$ as a percentage of the ADQW in the same case (full model). Grey: material parameters and linewidth; blue: triple-well growth errors (field re-tuned); orange: ADQW growth errors (0 V). Dashed line, 30% design gate; solid line, nominal.

cladding that serves as the ground contact. The TM_0 modes have $n_{\mathrm{eff}} = 3.231$ (10 μm) and 3.338 (5 μm), confinement factors $\Gamma_1 = 0.39$ and $\Gamma_2 = 0.51$, nonlinear overlap $|\int f_1^2 f_2^*|^2 = 0.053\,\mu\mathrm{m}^{-1}$, and background losses of 3.3 and 0.16 cm^{-1}. Because the 5 μm mode is much better confined, waveguide dispersion dominates and $L_{\mathrm{coh}} = 23.4\,\mu$m (23.2–25.2 μm including each medium's ISB index), much shorter than the 63 μm of bulk InP. The electrode period is therefore $\Lambda \approx 47\,\mu$m.

With the ground plane 5.31 μm below the electrodes, 30 V segments give a 56 kV/cm plateau in the stack (Fig. 4b). The field rounds over a few micrometres at segment boundaries, but because $\chi^{(2)}(F)$ saturates near its optimum the $\chi^{(2)}$ profile stays close to a square wave: the QPM coefficient is 92% of the ideal $(2/\pi)|\chi^{(2)}|_{\mathrm{max}}$ for 2 μm gaps (91–93% for 1–4 μm).

Because the linear response is even in F, the absorption and refractive index are the same under both electrodes and change only where the field passes through zero in the gaps, twice per period. Their modulation therefore has no Fourier component at the QPM wavevector $K = 2\pi/\Lambda$; its component at $2K$ is 27 cm^{-1} in the modal pump absorption and 3.1×10^{-3} in the modal pump index, for a plateau modal absorption of 238 cm^{-1}. For comparison, Stark on/off switching of the ADQW between 0 and -80 kV/cm (close to the best off fields at 5×10^{11} cm^{-2}, -55 to -85 kV/cm above 1 W, Sec. 4.6) writes a modal pump-absorption grating of 239 cm^{-1} and an index grating of 5.4×10^{-3} at K itself, and its on and off segments have different coherence lengths (25.2 and 28.9 μm at $n_s = 5 \times 10^{11}$ cm^{-2}).

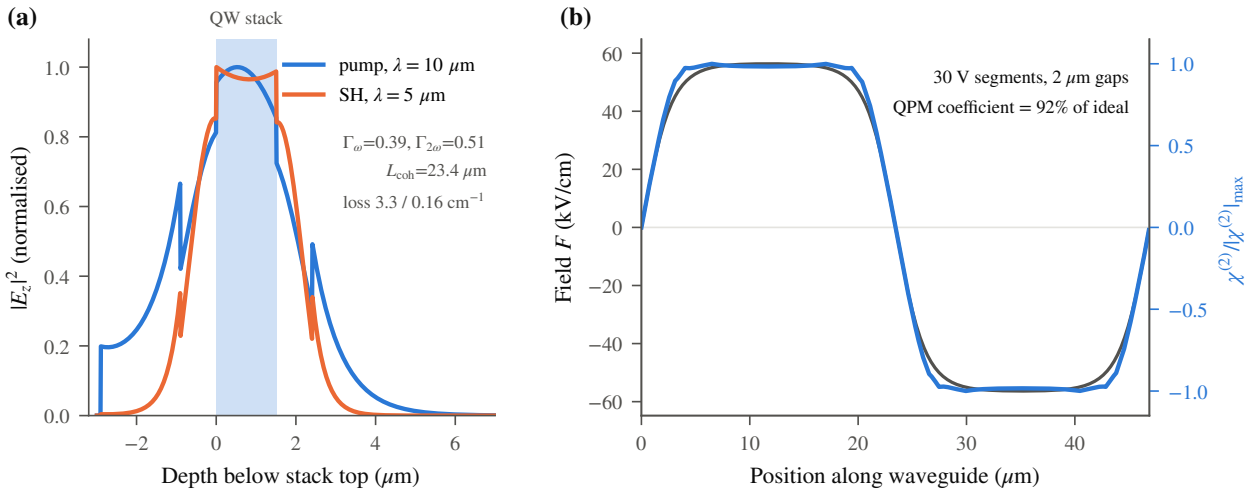


Figure 4: (a) $|E_z|^2$ of the TM_0 pump (10 μm) and SH (5 μm) modes; the shaded band is the QW stack. (b) Field at mid-stack along one electrode period and the resulting normalized $\chi^{(2)}(x)$ for 30 V segments with 2 μm gaps.

4.5 Relaxation and saturation

The LO-phonon rates of Eq. (7) give the ADQW's pump level a lifetime of 1.0 ps ($2 \to 1$) and its SH level 0.53 ps, close to the common 1 ps assumption. The triple well relaxes differently. Its third subband, the upper level of the pump transition, empties mainly into the second subband (0.78 ps), which is localized in the opposite outer well and returns to the ground state in 2.6 ps; at 55 kV/cm the second subband lies 50 meV above the ground state and already holds 17% of the electrons at equilibrium. Pumping therefore accumulates electrons in the second subband. Even so, the triple well's weaker pump transition (cross-section 2.53 times smaller than the ADQW's $1 \to 2$) makes it saturate at higher intensity: the small-signal pump absorption halves at 4.0 MW/cm^2 in the triple well and 2.0 MW/cm^2 in the ADQW (4.8 and 1.7 MW/cm^2 with a single 1 ps time), and the triple well's $|\chi^{(2)}|$ exceeds the ADQW's above a local pump intensity of 0.9–1.0 MW/cm^2 at all doping levels. In the waveguide the intensity in the stack is 1.74 MW/cm^2 per watt of pump power.

4.6 Conversion efficiency with saturation

Fig. 5 and Table 5 compare the electrically patterned schemes with growth-defined and idealized benchmarks, all with LO-phonon relaxation and a resonant 10 μm pump. Each value is the peak SH power along the waveguide, divided by the input pump power; the guide is up to 5 mm long for the benchmarks and, for the electrically written schemes, 6, 3.5, 1.5, 0.9 and 0.4 mm at $n_s = 10^{10}$, 3×10^{10}, 10^{11}, 2×10^{11} and 5×10^{11} cm^{-2}, beyond every optimum.

Triple well. Sign reversal is evaluated both at the design field of 55 kV/cm, where $|\chi^{(2)}|$ is largest, and with the bias amplitude optimized at each doping and power. Both are propagated through the electrode fields; the breakdown limit allows $F \leq 90$ kV/cm.

ADQW configurations.

- *Bipolar*: segments at $+F$ and $-F$, with $F \leq 90$ kV/cm optimized at each doping and power, through the electrode fields.
- *Stark on/off*: on at 0 V; off at the field, of either polarity, that maximizes the efficiency, through the electrode fields. Each segment has its own absorption, index and relaxation.
- *Idealized on/off*: a pattern that keeps the on-state absorption and index throughout.
- *Growth reversal*: ideal growth-defined orientation reversal, a lossless $\pm\chi^{(2)}$ square wave at 0 V.
- *No QPM*, and *perfect phase matching*, an unattainable bound.

Validity of the fields. Neighbouring electrodes differ by at most 188 kV/cm (breakdown limit, Sec. 3.3), and all fields are restricted to $|F| \leq 140$ kV/cm and to fields where no subband below $E_1 + 0.40$ eV is only partly localized in the active region. Beyond that, an upper subband anticrosses with states of the field-tilted spacer, which in a real stack belong to the neighbouring periods; a single-period model cannot describe these hybrids, and discarding them misplaces the 2ω resonance. An earlier version of this comparison took the Stark off field as the lowest field at which $|\chi^{(2)}|$ fell below 10% of its 0 V value. For a detuned pump that field was $+170$ to $+230$ kV/cm, where this happens, and the triple well was held at 55 kV/cm. Both choices biased the comparison.

Low pump power. At 10 mW growth reversal, which profits from the ADQW's larger small-signal $\chi^{(2)}$, is comparable to the triple well, and all electrical schemes lie within a factor of 1.3 of each other: the bias-optimized triple well reaches 0.89–1.28 times the bipolar ADQW, 0.95–1.13 times Stark QPM and 0.88–1.21 times growth reversal.

High pump power. As the power rises, saturation decides:

- With its bias optimized, the triple well gives 1.9–2.4 times Stark QPM at 1.26 W and 1.2–1.8 times at 10 W. It also gives 1.9–9.7 times the growth-reversed ADQW at 1.26–20 W. At its design field of 55 kV/cm, the triple well is 2.3–4.6 times less efficient at 10 W than with its bias optimized, because its doubly resonant ladder saturates most there.
- The bipolar ADQW is the most efficient scheme at every doping above 1 W: 1.4–2.4 times the bias-optimized triple well, reaching 2.3% at 10 W for $n_s = 5 \times 10^{11}$ cm^{-2}, above the perfectly phase-matched ADQW (0.31%).

Fringing. Relative to abrupt fields with no electrode factor, the electrode fields change the efficiency above 1 W by -44% to $+5\%$ for the triple well at 55 kV/cm, by -21% to $+38\%$ for the bipolar ADQW and by -17% to $+33\%$ for Stark QPM. A single electrode factor ($0.92^2 \approx 0.85$ in efficiency) therefore cannot represent all schemes.

Why the bipolar ADQW wins. Near resonance the sign of $\chi^{(2)}$ follows the sign of the pump detuning. A field of $\pm$55–90 kV/cm Stark-shifts the 1–2 transition of the ADQW by 16–33 meV to either side of the pump, while the 1–3 transition stays within 11 meV of $2\hbar\omega$. The two segments therefore have $\chi^{(2)}$ of nearly opposite phase: the phase difference is 151–189° at the optimum fields, with magnitudes of 22–55% of the 0 V value. Both segments are detuned, so they absorb and saturate less, the same effect that favours a detuned pump [44]. Mirror symmetry is therefore not needed for bias-defined phase reversal near resonance. What the triple well adds is an exact sign grating with identical segments.

The Stark on/off scheme also benefits from detuning: its best off segments are detuned rather than switched off. At $n_s = 5 \times 10^{11}$ cm^{-2} and above 1 W they lie at -55 to -85 kV/cm (at -140 kV/cm below 0.5 W), and at -70 kV/cm, the optimum at 10 W, the modal pump absorption falls from 838 to 112 cm^{-1} while $|\chi^{(2)}|$ keeps 36% of its 0 V value. It is weaker than the bipolar pattern because one of its segments stays resonant.

Sensitivity to the model. Table 6 tests the comparison at the design field against the model choices that matter most for saturation: linewidths of 6–14 meV, an additional 1 ps relaxation channel, and a single 1 ps relaxation time. The triple well's advantage at 1.26 and 10 W stays between 1.35 and 3.2 over growth reversal. Its ratio to Stark QPM stays between 1.26 and 2.31 at 1.26 W, and between 0.29 and 1.47 at 10 W; the Stark off field is held at its baseline optimum for these variations. The absolute efficiencies are much less robust: at 10 W they range from 0.11 to 0.75%, because they depend on how fast electrons return to the ground state. The ratios, not the absolute efficiencies, are the predictions of this work.

Detuned pumping. The comparison above uses a resonant 10 μm pump. At 10 W the resonant operating point lies beyond the conservative rate-equation limit (Rabi energy equal to the detuned linewidth $|D|$), which for these structures is near 4 W. A companion analysis of saturation limits [44] shows that a pump tuned 25–35 meV below resonance gives several times higher efficiency inside that limit. We therefore repeat the comparison with the pump red-detuned by up to 40 meV, at $n_s = 10^{11}$ and 5×10^{11} cm^{-2}. At each wavelength:

- the waveguide modes, confinement, overlap and background loss are recomputed;
- each medium's coherence length includes its own ISB index at ω and 2ω;
- the Stark off field is re-optimized among the valid fields at each photon energy.

Table 4 lists each scheme at its own best detuning. The resonant column of Table 5 is reproduced to within 0.2%. Detuning raises the efficiency of the triple well (by 1.4–5.7 times at its design field), of growth reversal and of phase matching. It also moves most optima to $|D| \gtrsim 2\gamma$, where the rate-equation limit (which scales as $|D|^2$) lies at 14–59 W; the optima at -10 meV have limits of 8–10 W. Four of the six Stark optima, including both at 10 W, stay at the resonant pump, where the limit is near 4 W, so the triple-well-to-Stark ratios at 10 W compare with points outside the validity of the rate equations.

With each scheme at its own best detuning:

- at 10 W the triple well gives 1.7–1.8 times the growth-reversed ADQW and 1.4–1.7 times Stark QPM;
- at 4 W it gives 1.06–1.26 and 2.0–2.3 times;

- at 1.26 W it is comparable to the growth-reversed ADQW (0.79–1.05 times) and gives 2.3 times Stark QPM.

The best Stark points stay closer to resonance (0 to -10 meV), because the off field already provides the detuning.

4.7 Optimized bias patterns

Table 5 restricts the bipolar ADQW to symmetric $\pm F$. To test whether either structure gains from more freedom, we also scan general two-field patterns (F_1, F_2), every ordered pair of valid fields on a 10 kV/cm grid up to ± 120 kV/cm (ADQW) or ± 100 kV/cm (triple well) within the breakdown limit, with the pump detuning scanned from 0 to -30 meV, each pattern propagated through its electrode fields. Table 3 lists the optima.

Triple well. Its best pattern is always symmetric, $\pm F$ with $F = 40$–80 kV/cm. Pure sign reversal is the best it can do.

ADQW. Its optimum is a broad plateau of opposite-polarity patterns at a resonant pump. On this 10 kV/cm grid the best symmetric pattern, ± 60 to ± 90 kV/cm, is within 1–6% of the best pair, $(+70, -50)$ to $(+100, -80)$ kV/cm, and the patterns are 1.3–1.4 times the best triple-well efficiency at 1.26 W and 1.8–2.2 times at 10 W (including the triple well's detuned points of Table 4). The ADQW optimum lies at the resonant pump, the edge of the scanned range; blue detuning was not considered.

Mechanism and prior art. The mechanism is the phase reversal by detuning described above. The 1–2 resonance is moved by 17–29 meV to either side of the pump, while the 1–3 resonance stays within 8 meV of $2\hbar\omega$, so the harmonic resonance survives; a red-detuned pump instead detunes 2ω by twice as much. Reference [18] used a uniform field on red-detuned transitions to control the resonant index mismatch, and Ref. [24] used Stark detuning to switch $\chi^{(2)}$ on and off in alternate segments; neither alternates the polarity from segment to segment to reverse the phase of $\chi^{(2)}$. The same detuning phase has been demonstrated directly in ISB polaritonic metasurfaces: in a coupled triple well with equal Stark coefficients for E_{21} and E_{32} the complex $\chi^{(2)}$ traces a cardioid as the bias is swept, and two meta-atoms with opposite field overlaps per unit cell give independent 0–2π phase and on/off amplitude control with two biases, used for electrically programmed second-harmonic phase and amplitude gratings [25]. That is local phase control in a 500-nm film, with the π offset supplied by the resonator geometry; here the detuning phase is used along a waveguide to quasi-phase-match.

Costs. Its costs relative to the triple well are:

- unequal segment magnitudes;
- segments with different coherence lengths (19.5–19.9 and 28.9–32.0 μm at $n_s = 5 \times 10^{11}$ cm^{-2});
- an absorption and index grating at the QPM period.

Table 3: Best periodic two-field bias patterns (F_1, F_2) in kV/cm, with both fields free among valid fields on a 10 kV/cm grid of up to ±120 kV/cm (ADQW) or ±100 kV/cm (triple well) and the pump detuning (meV) scanned from 0 to −30 meV in 10 meV steps (the triple well reaches 1.266% at 5×10^{11} cm^{-2} and 10 W with ±55 kV/cm at −25 meV, Table 4); peak conversion efficiency (%) with LO-phonon relaxation, both structures propagated through their electrode fields with neighbouring fields within 188 kV/cm.

n_s (cm^{-2})	Pump	ADQW: (F_1, F_2), detuning	ADQW (%)	TW: (F_1, F_2), detuning	TW (%)
10^{11}	1.26 W	$(+80, -60)$, 0	0.252	(± 50), −10	0.191
10^{11}	10 W	$(+100, -70)$, 0	1.825	(± 70), −10	0.842
5×10^{11}	1.26 W	$(+70, -50)$, 0	0.348	(± 40), −20	0.244
5×10^{11}	10 W	$(+100, -80)$, 0	2.287	(± 80), −10	1.234

Table 4: Peak conversion efficiency (%) with LO-phonon relaxation for each scheme at its own best red detuning of the pump (in meV, in parentheses; scanned 0 to −40 meV, Stark to −35 meV, with the waveguide recomputed at each wavelength; Stark off field optimized among valid fields at each detuning; the triple well and Stark QPM propagated through their electrode fields), and the ratios of the triple well to Stark QPM and to ideal growth reversal.

n_s (cm^{-2})	Pump	TW sign	Stark	rev.	PM	TW/Stark	TW/rev.
10^{11}	1.26 W	0.187 (−10)	0.080 (−10)	0.177 (−25)	0.433 (−25)	2.34	1.05
10^{11}	4 W	0.398 (−20)	0.202 (−10)	0.316 (−30)	0.768 (−30)	1.97	1.26
10^{11}	10 W	0.725 (−20)	0.505 (0)	0.393 (−30)	0.950 (−30)	1.44	1.84
5×10^{11}	1.26 W	0.231 (−20)	0.102 (0)	0.293 (−30)	0.704 (−30)	2.26	0.79
5×10^{11}	4 W	0.629 (−25)	0.279 (0)	0.594 (−35)	1.407 (−35)	2.26	1.06
5×10^{11}	10 W	1.266 (−25)	0.756 (0)	0.762 (−35)	1.785 (−35)	1.67	1.66

Its best patterns also need higher fields, 50–100 kV/cm against 40–80 kV/cm for the triple well.

The triple well's optimum detuning (−10 to −25 meV) is smaller than that of growth reversal and phase matching (−25 to −35 meV), and its efficiency falls steeply beyond −30 meV. Table 4 holds the triple well at 55 kV/cm.

5 Discussion

Relation to earlier proposals. Bias-defined sign reversal of a field-induced ISB $\chi^{(2)}$ is an old idea [29, 30]; what has been missing is a symmetric structure in which it is large. Up to 140 kV/cm, single, double and stepped symmetric wells give at most 34–40% of an optimized ADQW, and only at that field, close to the field at which their upper subbands unbind and tunnelling becomes significant. The coupled triple well reaches 55% at 55 kV/cm because its outer-well doublet converts a modest field into a full localization of the ground state, so that the structure becomes an ADQW of either orientation rather than a weakly perturbed symmetric well. The approach differs from voltage-controlled true phase matching [17], which tunes the resonant refractive index and keeps the sign of $\chi^{(2)}$ fixed, and from Stark on/off QPM [24], which modulates the magnitude and the linear response together. Bipolar Stark detuning of an ordinary ADQW, which reverses the phase of the resonant $\chi^{(2)}$ without mirror symmetry (Sec. 4.7), is related to the field-modulated detuned resonances of Ref. [18] and is the more efficient electrical pattern in our calculations.

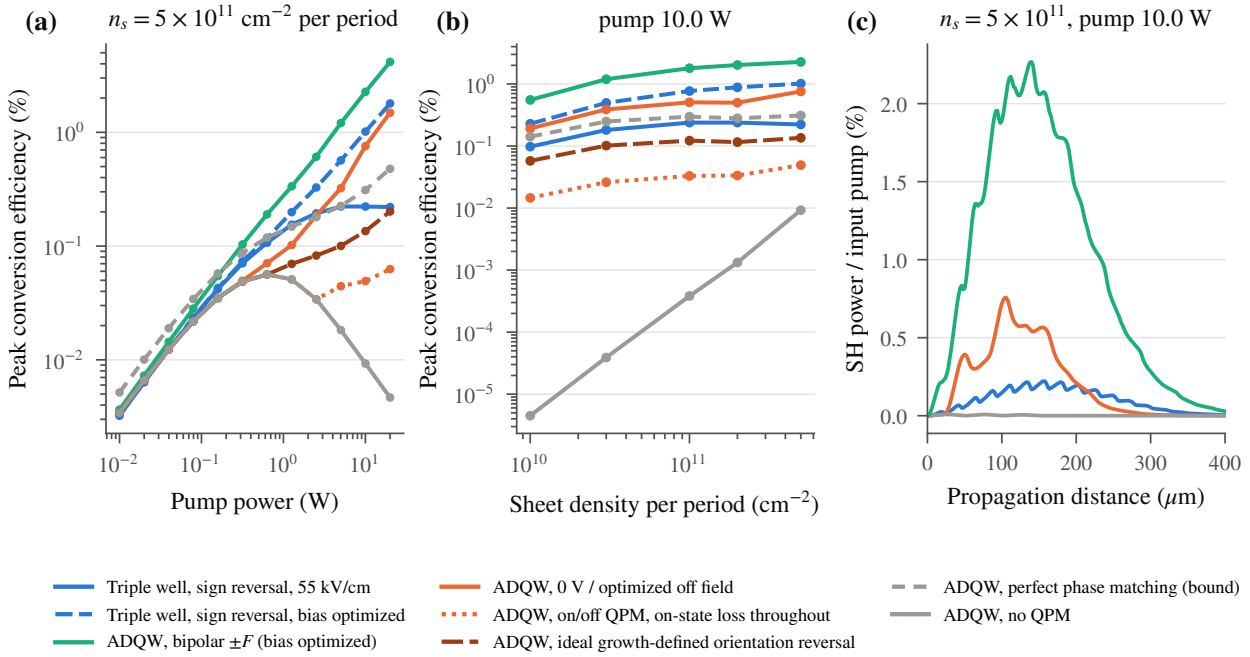


Figure 5: Saturated, pump-depleted SHG in the waveguide of Fig. 4, with LO-phonon relaxation and a resonant pump. (a) Peak conversion efficiency versus pump power at $n_s = 5 \times 10^{11}$ cm^{-2} per period. (b) Peak efficiency versus doping at 10 W. (c) SH power along the waveguide at 10 W and $n_s = 5 \times 10^{11}$ cm^{-2}. Schemes: the triple well at 55 kV/cm and with its bias optimized at each point; the ADQW with bipolar segments $\pm F$ (F optimized at each point); Stark modulation between 0 V and the best off field at each point (the off segment is detuned, not switched off); idealized on/off; ideal growth reversal; perfect phase matching; and no QPM. The electrical schemes are propagated through their electrode fields with neighbouring fields within 188 kV/cm; all fields are restricted to those where the single-period model is valid.

Where the scheme is useful. Pure sign reversal by bias polarity beats growth-defined orientation reversal where saturation limits conversion, at pump intensities above at most 0.02–0.6 MW/cm^2 in the stack, depending on doping. It is not the most efficient electrical pattern, however. An ordinary ADQW under bipolar bias also reverses the phase of $\chi^{(2)}$, by moving its resonance across the pump, and its detuned segments saturate less, so it converts 1.4–2.4 times as much at watt-level pumping. The triple well is the better choice when an exact, symmetric sign grating matters: identical segments with one coherence length and loss, no linear grating at the QPM period, and a period that does not depend on the operating point. These properties matter for broadband or tunable operation and for reconfigurable patterns. Because both electrode polarities produce the same linear response, every segment has the same coherence length and loss, so the electrode period and duty cycle do not depend on the bias, doping or pump power; in Stark on/off QPM the on and off segments differ in coherence length by up to 24% (64% for the optimized two-field ADQW patterns), and the pattern must be designed for one operating point. Because the period is set by lithography rather than growth, one wafer can carry many periods, and the QPM condition can be trimmed by temperature or by the bias amplitude, which tunes the ISB index.

Practical constraints. The stack needs about 30 V across 5.3 μm with the semi-insulating cladding used here at the 55 kV/cm design field. The bias-optimized patterns of Table 5 need up to 90 kV/cm, about 48 V,

Table 5: Peak conversion efficiency (%) with LO-phonon relaxation, and the corresponding length, for the triple well with sign reversal at 55 kV/cm (TW 55) and with the bias amplitude optimized (TW opt.), and the ADQW with bipolar segments $\pm F$ (F optimized; ADQW bip.), Stark on/off QPM (0 V and the best off field), idealized on/off QPM with on-state loss throughout, ideal growth-defined orientation reversal (rev.), no QPM, or perfect phase matching (PM). The electrical schemes (TW 55, TW opt., ADQW bip., Stark) are propagated through their electrode fields with neighbouring fields within 188 kV/cm; fields are restricted to those where the single-period model is valid; the TW length refers to TW 55.

Pump	n_s (cm^{-2})	TW 55	TW opt.	ADQW bip.	Stark	on/off (ideal.)	rev.	no QPM	PM	TW length
1.26 W	10^{10}	0.071	0.087	0.128	0.041	0.012	0.046	$< 10^{-4}$	0.114	2.25 mm
1.26 W	10^{11}	0.136	0.168	0.251	0.074	0.019	0.063	0.003	0.153	300 μm
1.26 W	5×10^{11}	0.154	0.199	0.335	0.102	0.051	0.070	0.051	0.148	64 μm
10.0 W	10^{10}	0.098	0.228	0.554	0.192	0.015	0.058	$< 10^{-4}$	0.142	3.77 mm
10.0 W	10^{11}	0.238	0.770	1.796	0.506	0.033	0.122	4×10^{-4}	0.298	653 μm
10.0 W	5×10^{11}	0.223	1.014	2.269	0.757	0.049	0.135	0.009	0.311	157 μm

Table 6: Sensitivity of the comparison to the linewidth Γ (HWHM) and the relaxation model: ratio of the triple well's peak efficiency to the growth-reversed ADQW (TW/rev.) and to Stark QPM (TW/Stark; off field fixed at its baseline optimum). Ranges are over $\Gamma = 6$, 10 and 14 meV with LO-phonon relaxation, and over LO-phonon relaxation plus a 1 ps channel and a single 1 ps time at $\Gamma = 10$ meV; electrode fields as in Table 5.

n_s (cm^{-2})	Pump	TW/rev.	TW/Stark	TW efficiency (%)
10^{11}	10 mW	1.07–1.16	1.00–1.09	0.001–0.008
10^{11}	1.26 W	1.61–2.67	1.47–2.10	0.06–0.39
10^{11}	10 W	1.43–2.10	0.43–1.47	0.11–0.75
5×10^{11}	10 mW	0.93–1.02	0.93–1.02	0.001–0.012
5×10^{11}	1.26 W	1.62–3.24	1.26–2.31	0.07–0.50
5×10^{11}	10 W	1.35–2.02	0.29–0.93	0.11–0.73

which puts up to 96 V across a $2\,\mu$m gap between electrodes of opposite polarity, within the 0.5 MV/cm breakdown limit we impose; the Stark off segments use up to 140 kV/cm (74 V) and the two-field patterns of Table 3 up to 100 kV/cm (53 V). At high power the bias-optimized patterns sit at that limit, so their efficiencies depend on the assumed breakdown field and gap. These fields approach the range above about 100–140 kV/cm where upper subbands hybridize with those of neighbouring periods, which a single-period model does not capture. Placing doped contact layers adjacent to the stack would reduce this to about 8 V at the cost of free-carrier loss. Thermionic escape over the 0.40 eV barrier gives an inter-period hopping time of about 0.4 μs at 300 K (a leakage of $\sim$0.2 A/cm^2). Because the semi-insulating cladding blocks this current, the escaping charge accumulates at the stack boundary and would screen a dc bias in roughly the same time. The bias should therefore be pulsed in step with the pump on sub-microsecond time scales, or the device cooled below about 200 K, where the hopping time exceeds a millisecond. Tunnelling escape of the resonant subbands is negligible at 55 kV/cm. The watt-level results imply strong heating: at 10 W and $n_s = 5 \times 10^{11}$ cm^{-2} the stack absorbs 6.7 W over the $157\,\mu$m optimum length, 1.9×10^9 W/cm^3, an adiabatic heating rate of about 1.3 K/ns. They therefore apply to pulses of about 10 ns or shorter at low duty cycle, or to correspondingly lower powers; continuous-wave operation would require a thermal design not attempted here. The steady-state rate equations are valid for such pulses, which are much longer than the picosecond

Table 7: Proposed layer sequence (one period; 40 periods), grown on n-InP.

Layer	Thickness	Notes
$In_{0.52}Al_{0.48}As$ spacer	100 Å	Si δ-sheet 2.5×10^{11} cm^{-2} at its outer edge
$In_{0.53}Ga_{0.47}As$ outer well	58 Å	
$In_{0.52}Al_{0.48}As$ barrier	14 Å	
$In_{0.53}Ga_{0.47}As$ centre well	34 Å	
$In_{0.52}Al_{0.48}As$ barrier	14 Å	
$In_{0.53}Ga_{0.47}As$ outer well	58 Å	
$In_{0.52}Al_{0.48}As$ spacer	100 Å	Si δ-sheet 2.5×10^{11} cm^{-2} at its outer edge

relaxation times. The outer-well width must be controlled to about a monolayer, which is within current molecular-beam and metal-organic epitaxy practice for QCLs.

Limitations of the model. We use a single-band model with literature parameters; Lorentzian broadening, which overestimates absorption far from resonance [6]; first-order depolarization without excitonic or local-field corrections [37, 38]; LO-phonon relaxation for electrons at the subband minima with bulk phonons, without interface-roughness or electron–electron scattering; a planar waveguide and a two-dimensional, uniform-permittivity electrode model without fringing at the ridge edges, and a bulk breakdown field for the gaps; linearly interpolated quaternary parameters for the stepped well; and assumed mid-infrared indices for the ternary alloys. These affect the triple well and the ADQW similarly, so the ratios we report are more robust than the absolute values, as the error budget and Table 6 confirm. The coherence length depends on the alloy dispersion and must be calibrated experimentally; this is why lithographic periods are attractive.

Extensions. The same symmetry argument applies to difference-frequency generation of terahertz radiation [7, 8], provided the nonlinear section is separate from a lasing QCL core, whose bias polarity cannot alternate. It also applies to other material systems with symmetric coupled wells, and to electrically patterned nonlinear metasurfaces [12, 25], where bias-programmed phase gratings have now been demonstrated [25].

Proposed experiment. A first test needs one 40-period InGaAs/InAlAs stack on n-InP with the layer sequence of Table 7. Three measurements would test the central predictions: (1) ISB absorption at $\pm F$, which should be identical; (2) SHG of a pulsed 10 μm pump through a uniform-bias section versus F, with bias pulses synchronized to the pump, which should be odd in F and peak near 55 kV/cm; and (3) SHG through segmented sections with a range of electrode periods around 47 μm, which should show a QPM resonance absent in sections without segmentation.

6 Conclusion

Quasi-phase matching of intersubband second-harmonic generation can be written electrically, by segmented electrodes of alternating polarity, and it competes with growth-defined orientation reversal and Stark on/off QPM once saturation limits the conversion. We analysed two routes with self-consistent electrostatics, depolarization, growth errors, a realistic waveguide, the fields of the actual electrodes with a breakdown limit, and LO-phonon-resolved saturable propagation.

- An ordinary ADQW under bipolar bias is the most efficient electrical scheme, 1.4–2.4 times as efficient as the bias-optimized triple well above 1 W, because a Stark shift across the pump reverses the phase of its resonant $\chi^{(2)}$ and detunes both segments. This applies the detuning-phase control demonstrated in intersubband metasurfaces [25] along a waveguide.

- A mirror-symmetric coupled triple well is an electrically reversible asymmetric quantum well that retains about half of an optimized ADQW's nonlinearity at 55 kV/cm. Its $\chi^{(2)}$ is exactly odd in the field and its linear response exactly even, so alternating-polarity electrodes write an exact sign grating with identical segments, no linear grating at the QPM period and a bias-independent period.

- With its bias optimized, the triple well outperforms growth-defined orientation reversal by 1.9–9.7 times above about 1 W and Stark on/off QPM by up to 2.4 times near 1 W.

Bias-defined phase reversal near resonance therefore does not require mirror symmetry. The bipolar ADQW is the choice for efficiency; the triple well is the choice where an exact, symmetric sign grating matters, and it gives a long-standing proposal for bias-defined QPM a concrete design. Both can be grown and tested with standard InP-based QCL technology, with pulsed bias and pump.

Data and code availability

The `qwkit` software package (v0.7.0) is archived at `https://doi.org/10.5281/zenodo.23444120`. The simulation scripts, raw JSON datasets, and figure-generation code that reproduce every result in this study are permanently archived at `https://doi.org/10.5281/zenodo.23444557`.

References

[1] M. M. Fejer, S. J. B. Yoo, R. L. Byer, A. Harwit, and J. S. Harris, Jr. Observation of extremely large quadratic susceptibility at 9.6–10.8 μm in electric-field-biased AlGaAs quantum wells. *Phys. Rev. Lett.*, 62:1041, 1989. doi: 10.1103/PhysRevLett.62.1041.

[2] E. Rosencher and Ph. Bois. Model system for optical nonlinearities: Asymmetric quantum wells. *Phys. Rev. B*, 44:11315, 1991. doi: 10.1103/PhysRevB.44.11315.

[3] E. Rosencher, A. Fiore, B. Vinter, V. Berger, Ph. Bois, and J. Nagle. Quantum engineering of optical nonlinearities. *Science*, 271:168, 1996. doi: 10.1126/science.271.5246.168.

[4] J. R. Meyer, C. A. Hoffman, F. J. Bartoli, E. R. Youngdale, and L. R. Ram-Mohan. Momentum-space reservoir for enhancement of intersubband second-harmonic generation. *IEEE J. Quantum Electron.*, 31:706–714, 1995.

[5] I. Vurgaftman, J. R. Meyer, and L. R. Ram-Mohan. Optimized second-harmonic generation in asymmetric double quantum wells. *IEEE J. Quantum Electron.*, 32:1334, 1996. doi: 10.1109/3.511546.

[6] I. Vurgaftman, M. P. Lumb, and J. R. Meyer. *Bands and Photons in III–V Semiconductor Quantum Structures*. Oxford University Press, Oxford, 2021. doi: 10.1093/oso/9780198767275.001.0001.

[7] M. A. Belkin, F. Capasso, A. Belyanin, D. L. Sivco, A. Y. Cho, D. C. Oakley, C. J. Vineis, and G. W. Turner. Terahertz quantum-cascade-laser source based on intracavity difference-frequency generation. *Nat. Photonics*, 1:288, 2007. doi: 10.1038/nphoton.2007.70.

[8] K. Fujita, S. Jung, Y. Jiang, J. H. Kim, A. Nakanishi, A. Ito, M. Hitaka, T. Edamura, and M. A. Belkin. Recent progress in terahertz difference-frequency quantum cascade laser sources. *Nanophotonics*, 7: 1795, 2018. doi: 10.1515/nanoph-2018-0093.

[9] J. Lee, M. Tymchenko, C. Argyropoulos, P.-Y. Chen, F. Lu, F. Demmerle, G. Boehm, M.-C. Amann, A. Alù, and M. A. Belkin. Giant nonlinear response from plasmonic metasurfaces coupled to intersubband transitions. *Nature*, 511:65, 2014. doi: 10.1038/nature13455.

[10] R. Sarma, J. Xu, D. de Ceglia, N. Nookala, L. Carletti, S. Campione, J. Klem, S. D. Gennaro, M. B. Sinclair, M. A. Belkin, and I. Brener. All-dielectric intersubband polaritonic metasurface with giant second-order nonlinear response. In *Conference on Lasers and Electro-Optics (CLEO: Applications and Technology)*, page JM1G.4. Optica Publishing Group, 2020. doi: 10.1364/CLEO_AT.2020.JM1G.4.

[11] J. Yu, S. Park, I. Hwang, D. Kim, F. Demmerle, G. Boehm, M.-C. Amann, M. A. Belkin, and J. Lee. Electrically tunable nonlinear polaritonic metasurface. *Nat. Photonics*, 16:72, 2022. doi: 10.1038/s41566-021-00923-7.

[12] J. Yu, S. Park, I. Hwang, G. Boehm, M. A. Belkin, and J. Lee. Broadband giant nonlinear response using electrically tunable polaritonic metasurfaces. *Nanophotonics*, 13:1131, 2024. doi: 10.1515/nanoph-2023-0682.

[13] J. H. Krakofsky, S. Stich, M. Cotrufo, A. Mekawy, S. Mann, A. Köninger, G. Böhm, A. Alù, and M. A. Belkin. Overcoming intensity saturation in second harmonic nonlinear intersubband polaritonic metasurfaces using two-level systems. In *Photonic and Phononic Properties of Engineered Nanostructures XIII*, volume 12431 of *Proc. SPIE*, page 1243108, 2023. doi: 10.1117/12.2650045.

[14] P. Boucaud, F. H. Julien, D. D. Yang, J.-M. Lourtioz, E. Rosencher, and P. Bois. Saturation of second-harmonic generation in GaAs–AlGaAs asymmetric quantum wells. *Opt. Lett.*, 16:199, 1991. doi: 10.1364/OL.16.000199.

[15] G. Chesi, V. Falcone, S. Calcaterra, M. Virgilio, and J. Frigerio. Modelling second harmonic generation at mid-infrared frequencies in waveguide integrated Ge/SiGe quantum wells. *Opt. Express*, 31:17098, 2023. doi: 10.1364/OE.483888.

[16] K. L. Vodopyanov, K. O'Neill, G. B. Serapiglia, C. C. Phillips, M. Hopkinson, I. Vurgaftman, and J. R. Meyer. Phase-matched second harmonic generation in asymmetric double quantum wells. *Appl. Phys. Lett.*, 72:2654, 1998. doi: 10.1063/1.121088.

[17] J. R. Meyer, C. A. Hoffman, F. J. Bartoli, and L. R. Ram-Mohan. Intersubband second-harmonic generation with voltage-controlled phase matching. *Appl. Phys. Lett.*, 67(5):608–610, 1995.

[18] I. Vurgaftman, J. R. Meyer, and L. R. Ram-Mohan. Detuned intersubband resonances for broadband field-modulated second-harmonic generation in a waveguide. *Solid State Commun.*, 100(9):663–667, 1996. doi: 10.1016/0038-1098(96)00493-0.

[19] I. Vurgaftman, J. R. Meyer, and L. R. Ram-Mohan. Waveguide-mode intersubband second-harmonic generation with separate active and phase-matching regions. In *Nonlinear Optics: Materials, Fundamentals and Applications*, page NThE.21. Optica Publishing Group, 1996. doi: 10.1364/NLO.1996.NThE.21.

[20] S. Janz, F. Chatenoud, and R. Normandin. Quasi-phase-matched second-harmonic generation from asymmetric coupled quantum wells. *Opt. Lett.*, 19:622, 1994. doi: 10.1364/OL.19.000622.

[21] A. Fiore, Y. Beaulieu, S. Janz, J. P. McCaffrey, Z. R. Wasilewski, and D.-X. Xu. Quasiphase matched surface emitting second harmonic generation in periodically reversed asymmetric GaAs/AlGaAs quantum well waveguide. *Appl. Phys. Lett.*, 70:2655, 1997. doi: 10.1063/1.118987.

[22] J.-P. Bouchard, S. Janz, D.-X. Xu, Z. R. Wasilewski, U. G. Akano, I. V. Mitchell, P. G. Piva, and M. Têtu. Quasi-phase-matching using quantum-well-intermixing-induced modulation of the nonlinear susceptibility of asymmetric quantum wells. In *Applications of Photonic Technology 4*, volume 4087 of *Proc. SPIE*, page 597, 2000. doi: 10.1117/12.406458.

[23] K. Zeaiter, D. C. Hutchings, R. M. Gwilliam, K. Moutzouris, S. Venugopal Rao, and M. Ebrahimzadeh. Quasi-phase-matched second-harmonic generation in a GaAs/AlAs superlattice waveguide by ion-implantation-induced intermixing. *Opt. Lett.*, 28:911, 2003. doi: 10.1364/OL.28.000911.

[24] M. A. Belkin, M. Troccoli, L. Diehl, F. Capasso, A. A. Belyanin, D. L. Sivco, and A. Y. Cho. Quasiphase matching of second-harmonic generation in quantum cascade lasers by Stark shift of electronic resonances. *Appl. Phys. Lett.*, 88:201108, 2006. doi: 10.1063/1.2203938.

[25] J. Yu, J. Kim, H. Chung, J. Son, G. Boehm, M. A. Belkin, and J. Lee. Full complex amplitude control of second-harmonic generation via electrically tunable intersubband polaritonic metasurfaces. *Sci. Adv.*, 11:eadw8852, 2025. doi: 10.1126/sciadv.adw8852.

[26] E. Timurdogan, C. V. Poulton, M. J. Byrd, and M. R. Watts. Electric field-induced second-order nonlinear optical effects in silicon waveguides. *Nat. Photonics*, 11:200, 2017. doi: 10.1038/nphoton.2017.14.

[27] R. Yanagimoto, B. A. Ash, M. M. Sohoni, M. M. Stein, Y. Zhao, F. Presutti, M. Jankowski, L. G. Wright, T. Onodera, et al. Programmable on-chip nonlinear photonics. *Nature*, 649:330, 2026. doi: 10.1038/s41586-025-09620-9.

[28] L. Tsang, D. Ahn, and S. L. Chuang. Electric field control of optical second-harmonic generation in a quantum well. *Appl. Phys. Lett.*, 52:697, 1988. doi: 10.1063/1.99350.

[29] M. M. Fejer and R. L. Byer. Quantum well optical electric field biased nonlinear method and apparatus. U.S. Patent 4,880,297 (Stanford University), November 1989.

[30] D. Delacourt, J.-P. Pocholle, and M. Papuchon. Optical frequency doubler using quantum well semiconductor structures. U.S. Patent 5,289,309 (Thomson-CSF), February 1994.

[31] D. J. BenDaniel and C. B. Duke. Space-charge effects on electron tunneling. *Phys. Rev.*, 152:683, 1966. doi: 10.1103/PhysRev.152.683.

[32] D. F. Nelson, R. C. Miller, and D. A. Kleinman. Band nonparabolicity effects in semiconductor quantum wells. *Phys. Rev. B*, 35:7770, 1987. doi: 10.1103/PhysRevB.35.7770.

[33] I. Vurgaftman, J. R. Meyer, and L. R. Ram-Mohan. Band parameters for III–V compound semiconductors and their alloys. *J. Appl. Phys.*, 89:5815, 2001. doi: 10.1063/1.1368156.

[34] P.-O. Löwdin. On the non-orthogonality problem connected with the use of atomic wave functions in the theory of molecules and crystals. *J. Chem. Phys.*, 18:365, 1950. doi: 10.1063/1.1747632.

[35] R. W. Boyd. *Nonlinear Optics*. Academic Press, Burlington, MA, 3rd edition, 2008.

[36] T. Ando, A. B. Fowler, and F. Stern. Electronic properties of two-dimensional systems. *Rev. Mod. Phys.*, 54:437, 1982. doi: 10.1103/RevModPhys.54.437.

[37] M. Załużny. Influence of the depolarization effect on second-harmonic generation in asymmetric quantum wells. *Phys. Rev. B*, 51:9757, 1995. doi: 10.1103/PhysRevB.51.9757.

[38] J. N. Heyman, K. Craig, B. Galdrikian, M. S. Sherwin, K. Campman, P. F. Hopkins, S. Fafard, and A. C. Gossard. Resonant harmonic generation and dynamic screening in a double quantum well. *Phys. Rev. Lett.*, 72:2183, 1994. doi: 10.1103/PhysRevLett.72.2183.

[39] G. D. Pettit and W. J. Turner. Refractive index of InP. *J. Appl. Phys.*, 36:2081, 1965. doi: 10.1063/1.1714410.

[40] M. A. Ordal, R. J. Bell, R. W. Alexander, Jr., L. L. Long, and M. R. Querry. Optical properties of fourteen metals in the infrared and far infrared: Al, Co, Cu, Au, Fe, Pb, Mo, Ni, Pd, Pt, Ag, Ti, V, and W. *Appl. Opt.*, 24:4493, 1985. doi: 10.1364/AO.24.004493.

[41] Gracen Ownby. Electrically programmable intersubband second-harmonic gratings: synthesis under saturation, fringing fields and tuning. Zenodo, October 2026. URL https://doi.org/10.5281/zenodo.23444591.

[42] R. Ferreira and G. Bastard. Evaluation of some scattering times for electrons in unbiased and biased single- and multiple-quantum-well structures. *Phys. Rev. B*, 40:1074, 1989. doi: 10.1103/PhysRevB.40.1074.

[43] Scattering times for electrons in unbiased and biased single and multiple quantum wells. nextnano++ online tutorial, https://www.nextnano.com/docu/nextnanoplus/tutorials/1D_electron_scattering_QW.html, 2026. Accessed 10 October.

[44] Gracen Ownby. Rabi-frequency bound and saturation limits of resonant intersubband second-harmonic generation in waveguides and thin films. Zenodo, October 2026. URL https://doi.org/10.5281/zenodo.23444562.